AD 2. AERODROMES

VABV AD 2.1 AERODROME LOCATION INDICATOR AND NAME

VABV - BHAVNAGAR AIRPORT, BHAVNAGAR

VABV AD 2.2 AERODROME GEOGRAPHICAL AND ADMINISTRATIVE DATA

1	Aerodrome reference point coordinates and its site	214516N 0721126E 252.95 DEG/407M from physical extremity of RWY 25	
2	Direction and distance of aerodrome reference point from the center of the city or town which the aerodrome serves	145 DEG, 9KM from Bhavnagar railway station.	
3	Aerodrome elevation and reference temperature	43 FT / 40.0 DEG C	
4	Magnetic variation, date of information and annual change	0.36 DEG W (2010) /0.0333 DEG E	
5	Name of aerodrome operator, address, telephone, telefax, e-mail address, AFS address, website (if available)	Airports Authority of India, Bhavnagar Airport, Bhavnagar - 364001	
		Telephone:	(91) 278 2212971 (91) 278 2203213 (91) 9427211500
		Fax:	(91) 2782203991 (91) 2782201840
		AFS:	VABVYDYX VABVYHYX
		Email:	apdbv@AAI.AERO
6	Types of traffic permitted (IFR/VFR)	IFR/VFR	
7	Remarks	1. Elevation in EGM08 2. REQUEST FOR DOMESTIC NON-SKED MOV MUST BE FORWARDED TO AIRPORT DIRECTOR AT LEAST 24 HR [72 HR IN CASE OF MOV ON SAT] IN ADVANCE.	

VABV AD 2.3 OPERATIONAL HOURS

1	Aerodrome Operator	Mon-Fri: 0400-1230 UTC (0930-1800 IST) Sat, Sun + Hol : Nil
2	Custom and immigration	NIL
3	Health and sanitation	NIL
4	AIS briefing office	As ATS
5	ATS reporting office (ARO)	As ATS
6	MET Briefing office	As ATS
7	Air Traffic Service	Consult NOTAM for current ATS HR.
8	Fuelling	NIL
9	Handling	AIASL, GFHS
10	Security	As ATS
11	De-icing	NIL

12	Remarks	1. Outside of ATS HR services are available O/R with 24 HR PN to AD. 2. The ATS approved hourly RWY traffic handling capacity is as follows: Maximum number of ARR and DEP - 12 (the minimum spacing between two successive arrivals shall be more than five minutes) Maximum number of arrivals only - 06 Maximum number of departures only - 10 3. Airport Arrival Rate [AAR]-03

VABV AD 2.4 HANDLING SERVICES AND FACILITIES

1	Cargo-handling facilities	Limited - Manual
2	Fuel and Oil types	JET A1
3	Fuelling facilities and capacity	M/S. BHARAT PETROLEUM CORPORATION LIMITED [BPCL] 30 KL STORAGE, 24 KL REFUELER CONTACT: 91-9586750502 EMAIL: INCHARGE.BHAVNAGAR[A]BSSPL.IN
4	De-icing facilities	NIL
5	Hangar space for visiting aircraft	NIL
6	Repair facilities for visiting aircraft	NIL
7	Remarks	NIL

VABV AD 2.5 PASSENGER FACILITIES

1	Hotel(s) at or in the vicinity of aerodrome	In the city
2	Restaurant(s) at or in the vicinity of aerodrome	At AD and in the city.
3	Transportation possibilities	Taxis and buses from AD.
4	Medical Facilities	First aid at AD. Hospitals in the city.
5	Bank and post office at or in the vicinity of aerodrome	Banks: In the city Post office: In the city.
6	Tourist office	NIL
7	Remarks	NIL

VABV AD 2.6 RESCUE AND FIRE FIGHTING SERVICES

1	Aerodrome category for fire fighting	Within ATS HR: CAT-6
2	Rescue equipment	Available as per category.
3	Capability for removal of disabled aircraft	1. The critical aircraft for the airport is B737. 2. Primary responsibility for removal of disabled aircraft rests with the concerned airline. 3. The airlines may contact ‘Aerodrome Coordinator for Disabled Aircraft Removal Operations’ for coordinating with the agencies having Disabled Aircraft Recovery Kit. 4. Contact Details of Aerodrome Coordinator for Disabled Aircraft Removal Operations: Airport Director Office: 0278-2212971; Mobile: 9427211500 Email: apdbv@aai.aero
4	Remarks	NIL

VABV AD 2.7 SEASONAL AVAILABILITY CLEARING

1	Type(s) of clearing equipment	NIL

2	Clearance priorities	NIL
3	Remarks	NIL

VABV AD 2.8 APRONS, TAXIWAYS AND CHECK LOCATIONS/POSITIONS DATA

1	Designation, surface and strength of aprons	Refer Aircraft Parking Docking Chart
2	Designation, width, surface and strength of taxiways	Refer Aircraft Parking Docking Chart
3	Location and elevation of altimeter checkpoints	Location: 214511.3N 0721102.1E Elevation: 30FT
4	Location of VOR checkpoints	On TWY A 214509.47N 0721103.20E; ELEV: 28.01837 FT Elevation in EGM08
5	Position of INS checkpoints	NIL
6	Remarks	1. Apron Dimensions: 108M X 76.1M. 2. Coordinates and Elevation of RWY Holding Position i) TWY A: 214509.229N 0721103.312/28.22 FT ii) TWY B: 214506.507N 0721056.330E/33.90 FT iii) TWY C: 214516.896N 0721123.627E/23.02M 3. Coordinates and Elevation of RWY/Taxiway Intersection i) TWY A: 214506.454N 0721104.530E/32.94 FT ii) TWY B: 214503.794N 0721057.511E/36.03 FT iii) TWY C: 214514.156N 0721124.822E/23.29 FT 4. Isolated Aircraft Parking Position: Dimension: 76M X 76 M, Shoulder: 7.5M, Surface: Concrete, Strength (PCN): 66/R/C/W/T, Critical aircraft: B737 (Power-in/Power-out), Markings: Edge and Centreline markings, Lightings: Blue Edge Lights 5. Gravity Model: EGM-08

VABV AD 2.9 SURFACE MOVEMENT GUIDANCE AND CONTROL SYSTEM AND MARKINGS

1	Use of aircraft stand identification signs, taxiway guidelines and visual docking/parking guidance system at aircraft stands	Taxiing guidance provided on R/T Guidelines at Apron.
2	Runway and taxiway markings and lights	RWY Markings: Designation, THR, TDZ, Centreline, Edge, End, Turn pad (RWY 25) RWY Lights: Edge, End, THR RWY25, WingBar RWY07, RWY Guard LGT configuration A at RWY HLDG PSN on TWY A, B, C, Turn pad lights (RWY 25) TWY Marking: Centreline, Holding Position, Edge TWY Lights: Edge Apron: Edge
3	Stop bars (if any)	NIL
4	Remarks	AN ILLUMINATED WIND DIRECTION INDICATOR IN THE FORM OF TRUNCATED CONE MADE OF FABRIC OF RED AND WHITE COLOUR IN BANDS HAVE BEEN INSTALLED WHICH IS LOCATED ADJACENT TO FIRE APPROACH ROAD, NORTH OF R25 AND EAST OF TWY A.

VABV AD 2.10 AERODROME OBSTACLES

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	BUILDING	214451.1N 0721034.5E	65 FT	NIL	HOUSE
25/TKOF 07/APCH	POLE	214458.5N 0721032.1E	83 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	BUILDING	214457.1N 0721031.3E	73 FT	NIL	HOUSE
25/TKOF 07/APCH	TREE	214457.1N 0721030.8E	90 FT	NIL	TREE
25/TKOF 07/APCH	FENCE	214454.8N 0721033.8E	58 FT	NIL	Airport Boundary Wall with Fencing on Top
25/TKOF 07/APCH	TREE	214452.5N 0721027.8E	78 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	BUILDING	214449.0N 0721025.3E	72 FT	NIL	HOUSE
25/TKOF 07/APCH	BUILDING	214450.1N 0721023.6E	72 FT	NIL	HOUSE
25/TKOF 07/APCH	BUILDING	214447.7N 0721023.6E	74 FT	NIL	HOUSE
25/TKOF 07/APCH	BUILDING	214445.7N 0721022.8E	77 FT	NIL	HOUSE
25/TKOF 07/APCH	BUILDING	214446.3N 0721026.3E	95 FT	NIL	HOUSE
25/TKOF 07/APCH	BUILDING	214448.8N 0721019.7E	76 FT	NIL	HOUSE
25/TKOF 07/APCH	POLE	214443.5N 0721019.6E	106 FT	NIL	ELECT.POLE
25/TKOF 07/APCH	POLE	214445.3N 0721016.5E	104 FT	NIL	ELECT.POLE
25/TKOF 07/APCH	POLE	214447.0N 0721013.9E	104 FT	NIL	ELECT.POLE
25/TKOF 07/APCH	POLE	214447.7N 0721013.2E	105 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214450.8N 0721010.4E	106 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214451.2N 0721006.8E	105 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214449.8N 0721006.3E	101 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	TREE	214452.3N 0721007.9E	139 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	TREE	214450.8N 0721009.2E	139 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	TREE	214446.4N 0721011.8E	139 FT	NIL	EUCALYPTUS TREE
25/TKOF 07/APCH	TREE	214443.9N 0721015.2E	120 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214443.3N 0721015.1E	89 FT	NIL	BUILDING
25/TKOF 07/APCH	BUILDING	214448.5N 0721004.8E	119 FT	NIL	BUILDING

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	TREE	214453.7N 0721008.5E	131 FT	NIL	TREE
25/TKOF 07/APCH	OTHER	214446.6N 0721014.0E	89 FT	NIL	MOBILE ROAD TRAFFIC (ROAD ELEV. 22.0 M + TRAFFIC HT. 5 M)
25/TKOF 07/APCH	OTHER	214443.5N 0721018.7E	86 FT	NIL	Mobile Road Traffic (Road Elev. 21.3 M. + 5.0 M. Traffic Ht.)
25/TKOF 07/APCH	OTHER	214454.6N 0721034.3E	58 FT	NIL	Rod Top Localizer Antenna (Antenna Top 16.8 M.)
25/TKOF 07/APCH	ANTENNA	214457.5N 0721033.0E	77 FT	NIL	Antenna On Localizer Hut (Hut Top - 18.8 M)
25/TKOF 07/APCH	FENCE	214457.1N 0721032.7E	64 FT	NIL	Airport Boundary Wall with Fencing on Top
25/TKOF 07/APCH	OTHER	214457.3N 0721033.6E	64 FT	NIL	Mobile Road Traffic (Road Elev. 14.5 M. + 5.0 M. Traffic Ht.)
25/TKOF 07/APCH	FENCE	214458.0N 0721032.3E	70 FT	NIL	Airport Boundary Wall with Fencing on Top
25/TKOF 07/APCH	BUILDING	214457.7N 0721032.1E	72 FT	NIL	Building
25/TKOF 07/APCH	TREE	214458.2N 0721032.6E	80 FT	NIL	Tree
25/TKOF 07/APCH	BUILDING	214458.2N 0721032.6E	78 FT	NIL	House
25/TKOF 07/APCH	BUILDING	214458.1N 0721031.7E	73 FT	NIL	Building
25/TKOF 07/APCH	POLE	214459.2N 0721033.8E	80 FT	NIL	ELECT.POLE
25/TKOF 07/APCH	POLE	214459.2N 0721032.7E	86 FT	NIL	ELECT.POLE
25/TKOF 07/APCH	POLE	214458.5N 0721031.1E	87 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214458.4N 0721029.7E	93 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214451.9N 0721036.8E	60 FT	NIL	BUILDING
25/TKOF 07/APCH	BUILDING	214451.7N 0721035.9E	60 FT	NIL	Building
25/TKOF 07/APCH	TREE	214451.5N 0721035.8E	78 FT	NIL	Group of Trees
25/TKOF 07/APCH	POLE	214451.4N 0721036.5E	71 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	TREE	214451.2N 0721036.6E	75 FT	NIL	Group of Trees

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	POLE	214452.0N 0721031.3E	73 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214451.8N 0721030.6E	59 FT	NIL	Building
25/TKOF 07/APCH	POLE	214451.2N 0721030.7E	74 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214451.0N 0721029.8E	76 FT	NIL	Electric Pole
25/TKOF 07/APCH	TREE	214449.6N 0721022.4E	93 FT	NIL	Group of Trees
25/TKOF 07/APCH	POLE	214446.9N 0721023.4E	99 FT	NIL	Electric Pole
25/TKOF 07/APCH	TREE	214446.3N 0721021.7E	94 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	214454.4N 0721014.5E	117 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214454.9N 0721012.8E	100 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	214452.6N 0721009.5E	111 FT	NIL	Building
25/TKOF 07/APCH	POLE	214452.9N 0721009.3E	112 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214452.8N 0721008.5E	111 FT	NIL	Electric Pole
25/TKOF 07/APCH	OTHER	214453.7N 0721007.3E	100 FT	NIL	Mobile Road Traffic (Road Elev. 25.6 M. + 5.0 M. Traffic Ht.)
25/TKOF 07/APCH	OTHER	214452.7N 0721008.0E	128 FT	NIL	Temple (Flag Top)
25/TKOF 07/APCH	POLE	214447.8N 0721012.5E	102 FT	NIL	Light Pole
25/TKOF 07/APCH	POLE	214449.4N 0721011.1E	103 FT	NIL	Light Pole
25/TKOF 07/APCH	BUILDING	214449.6N 0721012.1E	101 FT	NIL	Building
25/TKOF 07/APCH	POLE	214449.2N 0721011.9E	105 FT	NIL	Electric Pole
25/TKOF 07/APCH	TREE	214448.7N 0721011.0E	118 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	214448.0N 0721011.7E	120 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214448.3N 0721013.3E	102 FT	NIL	BUILDING
25/TKOF 07/APCH	POLE	214445.3N 0721015.6E	98 FT	NIL	Light Pole
25/TKOF 07/APCH	POLE	214444.4N 0721018.1E	100 FT	NIL	Electric Pole
25/TKOF 07/APCH	TREE	214441.2N 0721013.0E	119 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	TREE	214444.0N 0721011.9E	120 FT	NIL	Group of Trees

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	OTHER	214434.2N 0720932.0E	211 FT	NIL	Cell Phone Mast
25/TKOF 07/APCH	BUILDING	214450.3N 0721005.0E	120 FT	NIL	Building
25/TKOF 07/APCH	POLE	214457.5N 0721031.4E	84 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214456.9N 0721029.9E	94 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214457.4N 0721029.6E	88 FT	NIL	Building
25/TKOF 07/APCH	TANK	214440.9N 0720925.4E	178 FT	NIL	Over Head Water Tank (Rod Top)
25/TKOF 07/APCH	OTHER	214457.9N 0721032.4E	77 FT	NIL	Security Hut
25/TKOF 07/APCH	FENCE	214458.7N 0721034.0E	67 FT	NIL	Airport Boundary Wall with Fencing on Top
25/TKOF 07/APCH	BUILDING	214430.8N 0720945.2E	164 FT	NIL	BUILDING
25/TKOF 07/APCH	BUILDING	214444.2N 0720955.3E	130 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	214439.9N 0720955.5E	151 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	214437.2N 0720958.1E	138 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	214436.0N 0720959.8E	135 FT	NIL	BUILDING
25/TKOF 07/APCH	BUILDING	214447.4N 0721005.2E	117 FT	NIL	BUILDING
25/TKOF 07/APCH	POLE	214452.0N 0721008.5E	117 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214452.2N 0721009.2E	114 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214450.2N 0721010.0E	111 FT	NIL	Electric Pole
25/TKOF 07/APCH	TREE	214441.9N 0721011.0E	123 FT	NIL	Group of Trees
25/TKOF 07/APCH	TREE	214449.6N 0721011.3E	124 FT	NIL	Group of Trees
25/TKOF 07/APCH	POLE	214449.9N 0721011.4E	105 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214448.3N 0721011.6E	108 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	BUILDING	214447.1N 0721011.8E	104 FT	NIL	House
25/TKOF 07/APCH	BUILDING	214453.6N 0721012.1E	115 FT	NIL	Building
25/TKOF 07/APCH	POLE	214447.4N 0721012.4E	109 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	TREE	214452.5N 0721013.0E	119 FT	NIL	Group of Trees

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	BUILDING	214450.9N 0721013.1E	106 FT	NIL	Temple
25/TKOF 07/APCH	TREE	214447.0N 0721014.0E	115 FT	NIL	Group of Trees
25/TKOF 07/APCH	POLE	214444.8N 0721015.7E	104 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	BUILDING	214449.8N 0721016.8E	97 FT	NIL	Building
25/TKOF 07/APCH	POLE	214444.1N 0721017.0E	104 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214452.7N 0721017.0E	112 FT	NIL	School Building
25/TKOF 07/APCH	TREE	214453.9N 0721017.9E	117 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214453.5N 0721018.2E	102 FT	NIL	House
25/TKOF 07/APCH	BUILDING	214442.4N 0721018.5E	107 FT	NIL	Building
25/TKOF 07/APCH	TREE	214442.8N 0721018.7E	115 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214445.1N 0721021.6E	101 FT	NIL	Temple
25/TKOF 07/APCH	BUILDING	214456.3N 0721021.6E	90 FT	NIL	HOUSE
25/TKOF 07/APCH	TANK	214450.2N 0721022.8E	94 FT	NIL	Sintex Tank on Building (Building Top 28.8 M.)
25/TKOF 07/APCH	BUILDING	214456.1N 0721024.0E	89 FT	NIL	House
25/TKOF 07/APCH	POLE	214456.9N 0721024.6E	83 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214447.7N 0721024.9E	94 FT	NIL	HOUSE
25/TKOF 07/APCH	POLE	214456.1N 0721025.7E	83 FT	NIL	Electric Pole
25/TKOF 07/APCH	TREE	214446.8N 0721026.0E	99 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214457.5N 0721025.9E	88 FT	NIL	House
25/TKOF 07/APCH	POLE	214451.9N 0721026.1E	84 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214454.6N 0721026.1E	83 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214453.4N 0721026.5E	83 FT	NIL	Electric Pole
25/TKOF 07/APCH	OTHER	214452.6N 0721026.6E	93 FT	NIL	Mast
25/TKOF 07/APCH	POLE	214446.4N 0721026.8E	94 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214452.6N 0721026.7E	83 FT	NIL	ELECTRIC POLE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	POLE	214456.2N 0721026.9E	82 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214451.8N 0721027.0E	83 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214450.0N 0721027.5E	83 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214454.3N 0721027.5E	81 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	BUILDING	214453.8N 0721027.5E	77 FT	NIL	House
25/TKOF 07/APCH	POLE	214453.6N 0721027.7E	81 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	BUILDING	214451.9N 0721027.8E	89 FT	NIL	House
25/TKOF 07/APCH	POLE	214448.2N 0721027.9E	84 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	TREE	214458.0N 0721027.7E	97 FT	NIL	Group of Trees
25/TKOF 07/APCH	POLE	214447.3N 0721028.2E	84 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214452.1N 0721028.1E	82 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214451.6N 0721028.2E	81 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	TREE	214455.3N 0721028.2E	93 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	TANK	214452.5N 0721028.5E	82 FT	NIL	Sintex Tank on Building (Building Top 25.6 M.)
25/TKOF 07/APCH	BUILDING	214448.2N 0721028.8E	86 FT	NIL	House
25/TKOF 07/APCH	POLE	214456.0N 0721028.9E	78 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	TREE	214450.4N 0721029.8E	90 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	POLE	214456.2N 0721029.9E	80 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214457.7N 0721030.2E	85 FT	NIL	HOUSE
25/TKOF 07/APCH	BUILDING	214456.2N 0721030.2E	72 FT	NIL	House
25/TKOF 07/APCH	TREE	214455.3N 0721030.6E	80 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	POLE	214449.9N 0721031.0E	76 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	BUILDING	214458.2N 0721030.9E	86 FT	NIL	HOUSE
25/TKOF 07/APCH	POLE	214449.1N 0721031.2E	76 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214450.7N 0721031.6E	75 FT	NIL	Electric Pole

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	TREE	214458.4N 0721032.1E	96 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214458.1N 0721032.1E	76 FT	NIL	HOUSE
25/TKOF 07/APCH	POLE	214451.7N 0721032.3E	73 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	BUILDING	214449.5N 0721032.5E	82 FT	NIL	House
25/TKOF 07/APCH	TREE	214457.0N 0721032.4E	84 FT	NIL	GROUP OF TREES
25/TKOF 07/APCH	BUILDING	214449.7N 0721032.6E	81 FT	NIL	HOUSE
25/TKOF 07/APCH	BUILDING	214459.1N 0721032.4E	88 FT	NIL	House
25/TKOF 07/APCH	POLE	214452.0N 0721033.0E	73 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214449.6N 0721033.5E	74 FT	NIL	House
25/TKOF 07/APCH	POLE	214452.8N 0721033.4E	71 FT	NIL	Electric Pole
25/TKOF 07/APCH	TREE	214450.6N 0721033.5E	83 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214459.6N 0721033.4E	81 FT	NIL	House
25/TKOF 07/APCH	POLE	214452.1N 0721033.7E	72 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214451.3N 0721034.1E	73 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214450.7N 0721034.4E	73 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214449.7N 0721034.5E	73 FT	NIL	HOUSE
25/TKOF 07/APCH	POLE	214450.1N 0721034.7E	73 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214451.2N 0721035.3E	72 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	TREE	214450.9N 0721035.7E	83 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214451.0N 0721036.3E	68 FT	NIL	HOUSE
25/TKOF 07/APCH	BUILDING	214458.8N 0721031.6E	87 FT	NIL	Building
25/TKOF 07/APCH	POLE	214458.7N 0721033.9E	80 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214458.5N 0721033.0E	83 FT	NIL	ELECTRIC POLE
25/TKOF 07/APCH	POLE	214458.2N 0721030.5E	88 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214457.6N 0721031.0E	86 FT	NIL	Building
25/TKOF 07/APCH	OTHER	214457.5N 0721033.0E	62 FT	NIL	Localizer Hut

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	BUILDING	214456.6N 0721022.6E	100 FT	NIL	Building
25/TKOF 07/APCH	TREE	214455.8N 0721030.1E	88 FT	NIL	Group of Trees
25/TKOF 07/APCH	OTHER	214455.3N 0721034.0E	58 FT	NIL	Rod On Localizer
25/TKOF 07/APCH	OTHER	214455.3N 0721033.9E	58 FT	NIL	L/C Rod Near Localizer
25/TKOF 07/APCH	BUILDING	214455.0N 0721020.7E	93 FT	NIL	Building
25/TKOF 07/APCH	NAVAID	214454.9N 0721034.1E	55 FT	NIL	Localizer Antenna
25/APCH 07/TKOF	BUILDING	214454.8N 0721027.9E	77 FT	NIL	Building
25/TKOF 07/APCH	POLE	214454.9N 0721034.1E	57 FT	NIL	Pole
25/TKOF 07/APCH	OTHER	214454.9N 0721033.9E	63 FT	NIL	Mobile Road Traffic (Road Elev. 14.3 M. + 5.0 M. Traffic Ht.)
25/TKOF 07/APCH	OTHER	214454.8N 0721033.8E	59 FT	NIL	Obstruction Light
25/TKOF 07/APCH	OTHER	214454.5N 0721034.2E	58 FT	NIL	L/C Rod Near Localizer
25/TKOF 07/APCH	POLE	214454.3N 0721022.5E	89 FT	NIL	Sintex Tank on Building (Building Top 26.1 M
25/TKOF 07/APCH	BUILDING	214453.0N 0721004.9E	144 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	214453.4N 0721028.4E	77 FT	NIL	Building
25/TKOF 07/APCH	TANK	214453.2N 0721020.4E	99 FT	NIL	Sintex Tank on Building (Building Top 28.5 M.)
25/TKOF 07/APCH	TREE	214453.2N 0721024.9E	104 FT	NIL	GROUP OF TREE
25/TKOF 07/APCH	POLE	214453.1N 0721024.5E	77 FT	NIL	Electric Pole
25/TKOF 07/APCH	FENCE	214453.2N 0721034.5E	58 FT	NIL	Airport Boundary wall with Fencing on Top
25/TKOF 07/APCH	OTHER	214452.8N 0721035.5E	64 FT	NIL	Mobile Road Traffic (Road Elev. 14.4 M. + 5.0 M. Traffic Ht)
25/TKOF 07/APCH	OTHER	214452.6N 0721024.6E	77 FT	NIL	Cremation Shed
25/TKOF 07/APCH	BUILDING	214452.5N 0721018.0E	99 FT	NIL	Building
25/TKOF 07/APCH	FENCE	214452.5N 0721034.9E	59 FT	NIL	Airport Boundary Wall with Fencing on Top

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	POLE	214452.5N 0721035.8E	53 FT	NIL	Light Pole
25/TKOF 07/APCH	POLE	214452.1N 0721020.5E	82 FT	NIL	Electric Pole
25/TKOF 07/APCH	OTHER	214451.8N 0721009.0E	98 FT	NIL	Mobile Road Traffic (Road Elev. 24.8 M. + 5.0 M. Traffic Ht.)
25/TKOF 07/APCH	TREE	214452.0N 0721025.8E	90 FT	NIL	Group of Trees
25/TKOF 07/APCH	POLE	214451.9N 0721025.2E	84 FT	NIL	Light Pole
25/TKOF 07/APCH	TANK	214451.6N 0721010.7E	112 FT	NIL	Sintex Tank on Building (Building Top 32.8 M.)
25/TKOF 07/APCH	GATE	214451.9N 0721035.1E	56 FT	NIL	Crash Gate
25/TKOF 07/APCH	BUILDING	214451.7N 0721021.0E	78 FT	NIL	Temple
25/TKOF 07/APCH	TREE	214451.7N 0721030.1E	89 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214451.3N 0721017.2E	94 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	214451.3N 0721021.9E	74 FT	NIL	Building
25/TKOF 07/APCH	TREE	214451.3N 0721019.5E	102 FT	NIL	Group of Trees
25/TKOF 07/APCH	POLE	214451.3N 0721025.3E	84 FT	NIL	Light Pole
25/TKOF 07/APCH	POLE	214451.1N 0721018.3E	86 FT	NIL	Electric Pole
25/TKOF 07/APCH	WALL	214450.9N 0721006.9E	101 FT	NIL	Mobile Road Traffic (Road Elev. 25.7 M. + 5.0 M. Traffic Ht.)
25/TKOF 07/APCH	POLE	214451.1N 0721026.3E	83 FT	NIL	Light Pole
25/TKOF 07/APCH	TREE	214451.1N 0721022.8E	92 FT	NIL	Group of Trees
25/TKOF 07/APCH	POLE	214451.0N 0721033.2E	74 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214450.8N 0721023.7E	72 FT	NIL	Building
25/TKOF 07/APCH	TREE	214450.7N 0721025.4E	98 FT	NIL	Group of Trees
25/TKOF 07/APCH	POLE	214450.5N 0721020.4E	83 FT	NIL	Electric Pole
25/TKOF 07/APCH	TANK	214450.3N 0721017.5E	99 FT	NIL	Sintex Tank on Building (Building Top 28.8 M.)
25/TKOF 07/APCH	POLE	214450.6N 0721032.6E	76 FT	NIL	Electric Pole

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	POLE	214450.0N 0721024.1E	80 FT	NIL	Electric Pole
25/TKOF 07/APCH	POLE	214450.0N 0721022.3E	84 FT	NIL	Electric Pole
25/TKOF 07/APCH	NAVAID	214450.0N 0721025.4E	90 FT	NIL	Building
25/TKOF 07/APCH	TANK	214449.9N 0721022.1E	97 FT	NIL	Sintex Tank on Building (Building Top 28.3 M.)
25/TKOF 07/APCH	BUILDING	214449.7N 0721024.2E	93 FT	NIL	Building
25/TKOF 07/APCH	TREE	214449.4N 0721023.4E	101 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214448.4N 0720958.8E	142 FT	NIL	Building
25/TKOF 07/APCH	WALL	214448.7N 0721021.8E	101 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214448.5N 0721015.8E	96 FT	NIL	Temple
25/TKOF 07/APCH	BUILDING	214448.8N 0721029.6E	88 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	214448.2N 0721020.2E	91 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	214447.8N 0721023.0E	81 FT	NIL	Building
25/TKOF 07/APCH	TANK	214447.7N 0721020.6E	99 FT	NIL	Sintex Tank on Building (Building Top 29.2 M.)
25/TKOF 07/APCH	TANK	214447.5N 0721023.8E	93 FT	NIL	Sintex Tank on Building (Building Top 27.6 M.)
25/TKOF 07/APCH	OTHER	214447.0N 0721013.3E	91 FT	NIL	Mobile Road Traffic (Road Elev. 22.7 M. + 5.0 M. Traffic Ht.)
25/TKOF 07/APCH	TANK	214446.6N 0721003.2E	122 FT	NIL	Sintex Tank on Building (Building Top 35.7 M.)
25/TKOF 07/APCH	TREE	214446.2N 0721022.3E	105 FT	NIL	Group of Trees
25/TKOF 07/APCH	BUILDING	214445.9N 0721021.2E	90 FT	NIL	Building
25/TKOF 07/APCH	POLE	214445.5N 0721014.5E	106 FT	NIL	Electric Pole
25/TKOF 07/APCH	TANK	214445.5N 0721014.1E	100 FT	NIL	Sintex Tank on Building (Building Top 29.7 M.)
25/TKOF 07/APCH	TANK	214445.3N 0721024.7E	94 FT	NIL	Sintex Tank on Building (Building Top 27.2 M.)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	POLE	214444.9N 0721005.5E	113 FT	NIL	Light Pole
25/TKOF 07/APCH	POLE	214444.7N 0721005.1E	106 FT	NIL	Electric Pole
25/TKOF 07/APCH	TANK	214444.6N 0721001.3E	128 FT	NIL	Sintex Tank on Building (Building Top 36.6 M.)
25/TKOF 07/APCH	OTHER	214444.7N 0721016.6E	139 FT	NIL	Mast
25/TKOF 07/APCH	TREE	214444.5N 0721021.8E	120 FT	NIL	Group of Trees
25/TKOF 07/APCH	OTHER	214443.5N 0720937.1E	182 FT	NIL	Mast
25/TKOF 07/APCH	POLE	214443.9N 0721017.2E	105 FT	NIL	Electric Pole
25/TKOF 07/APCH	BUILDING	214443.9N 0721020.7E	97 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	214442.9N 0721017.2E	105 FT	NIL	Building
25/TKOF 07/APCH	POLE	214442.7N 0721005.0E	111 FT	NIL	Light Pole
25/TKOF 07/APCH	BUILDING	214441.9N 0720949.6E	142 FT	NIL	Group of Buildings (U/C)
25/TKOF 07/APCH	BUILDING	214441.9N 0720955.4E	157 FT	NIL	Mast
25/TKOF 07/APCH	TANK	214442.1N 0721001.6E	124 FT	NIL	Sintex Tank on Building (Building Top 37.2 M.)
25/TKOF 07/APCH	OTHER	214442.2N 0721021.0E	88 FT	NIL	Mobile Road Traffic (Road Elev. 21.8 M. + 5.0 M. Traffic Ht.)
25/TKOF 07/APCH	TANK	214441.6N 0721003.7E	125 FT	NIL	Sintex Tank on Building (Building Top 36.7 M.)
25/TKOF 07/APCH	POLE	214441.4N 0721005.0E	110 FT	NIL	Light Pole
25/TKOF 07/APCH	BUILDING	214441.1N 0721006.5E	124 FT	NIL	Building (U/C)
25/TKOF 07/APCH	BUILDING	214441.0N 0721010.7E	132 FT	NIL	Building (U/C)
25/TKOF 07/APCH	BUILDING	214439.9N 0720951.8E	156 FT	NIL	Building
25/TKOF 07/APCH	BUILDING	214440.2N 0721010.9E	125 FT	NIL	Building (U/C)
25/TKOF 07/APCH	BUILDING	214438.7N 0720948.6E	140 FT	NIL	Group Of Buildings (U/C)
25/TKOF 07/APCH	OTHER	214437.2N 0720932.7E	176 FT	NIL	Mast
25/TKOF 07/APCH	BUILDING	214435.1N 0721005.3E	136 FT	NIL	Building

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
25/TKOF 07/APCH	TANK	214432.1N 0720946.3E	156 FT	NIL	Sintex Tank on Building (Building Top 45.7 M.)
25/TKOF 07/APCH	BUILDING	214431.5N 0720945.7E	162 FT	NIL	Group Of Buildings (U/C)
25/TKOF 07/APCH	OTHER	214426.0N 0720933.7E	179 FT	NIL	Mast
25/APCH 07/TKOF	OTHER	214523.7N 0721142.5E	36 FT	NIL	Mobile Road Traffic (Road Elev. 6.0 M. + 5.0 M. Traffic Ht.)
25/APCH 07/TKOF	FENCE	214523.4N 0721141.1E	27 FT	NIL	Airport Boundary Wall with Fencing On Top
25/APCH 07/TKOF	POLE	214523.1N 0721141.2E	25 FT	NIL	Light Pole
25/APCH 07/TKOF	OTHER	214519.0N 0721145.5E	36 FT	NIL	Mobile Road Traffic (Road Elev. 6.0 M. + 5.0 M. Traffic Ht.)
25/APCH 07/TKOF	POLE	214518.5N 0721144.0E	25 FT	NIL	Light Pole
25/APCH 07/TKOF	FENCE	214517.9N 0721143.5E	27 FT	NIL	Airport Boundary Wall with Fencing on Top
25/APCH 07/TKOF	OTHER	214521.8N 0721144.9E	36 FT	NIL	Mobile Road Traffic (Road Elev. 6.1 M. + 5.0 M. Traffic Ht.)
In circling area and at AD	BUILDING	214452.2N 0721037.5E	63 FT	NIL	HOUSE
In circling area and at AD	BUILDING	214459.1N 0721035.1E	63 FT	NIL	HOUSE
In circling area and at AD	POLE	214459.6N 0721034.6E	77 FT	NIL	ELECT.POLE
In circling area and at AD	POLE	214445.2N 0721027.0E	96 FT	NIL	ELECT.POLE
In circling area and at AD	POLE	214454.0N 0721007.6E	119 FT	NIL	ELECT.POLE
In circling area and at AD	BUILDING	214440.1N 0721020.9E	108 FT	NIL	Building
In circling area and at AD	TREE	214459.2N 0721034.2E	84 FT	NIL	Group of Trees
In circling area and at AD	TREE	214459.5N 0721032.5E	92 FT	NIL	Tree Top
In circling area and at AD	POLE	214459.8N 0721032.3E	86 FT	NIL	ELECT.POLE
In circling area and at AD	TREE	214459.9N 0721032.1E	96 FT	NIL	Group of Trees
In circling area and at AD	POLE	214459.6N 0721031.8E	87 FT	NIL	ELECT.POLE
In circling area and at AD	POLE	214451.3N 0721037.7E	71 FT	NIL	ELECT.POLE

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TREE	214455.9N 0721012.7E	119 FT	NIL	Group of Trees
In circling area and at AD	TREE	214457.1N 0721013.6E	125 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	214454.5N 0721006.0E	141 FT	NIL	GROUP OF TREES
In circling area and at AD	OTHER	214453.0N 0721038.1E	63 FT	NIL	Mobile Road Traffic (Road Elev. 14.1 M. + 5.0 M. Traffic Ht.)
In circling area and at AD	TREE	214456.3N 0721015.5E	131 FT	NIL	GROUP OF TREES
In circling area and at AD	TREE	214443.9N 0721026.6E	105 FT	NIL	Group of Trees
In circling area and at AD	BUILDING	214459.1N 0721027.0E	92 FT	NIL	HOUSE
In circling area and at AD	BUILDING	214459.6N 0721028.7E	90 FT	NIL	HOUSE
In circling area and at AD	BUILDING	214458.7N 0721034.0E	71 FT	NIL	HOUSE
In circling area and at AD	BUILDING	214500.0N 0721034.0E	77 FT	NIL	HOUSE
In circling area and at AD	POLE	214500.3N 0721034.4E	77 FT	NIL	ELECTRIC POLE
In circling area and at AD	POLE	214450.6N 0721036.7E	72 FT	NIL	ELECTRIC POLE
In circling area and at AD	POLE	214449.5N 0721037.3E	73 FT	NIL	ELECTRIC POLE
In circling area and at AD	BUILDING	214449.8N 0721037.3E	71 FT	NIL	HOUSE
In circling area and at AD	TREE	214452.4N 0721037.7E	79 FT	NIL	Group of Trees
In circling area and at AD	TREE	214451.0N 0721037.9E	87 FT	NIL	GROUP OF TREES
In circling area and at AD	OTHER	214451.0N 0721038.3E	69 FT	NIL	Shed
In circling area and at AD	BUILDING	214504.7N 0721049.5E	56 FT	NIL	Temple
In circling area and at AD	FENCE	214503.9N 0721049.3E	43 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	POLE	214503.3N 0721048.0E	41 FT	NIL	Light Pole
In circling area and at AD	FENCE	214502.4N 0721043.9E	48 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	TANK	214500.7N 0721026.6E	96 FT	NIL	Sintex Tank on Building (Building Top 28.3 M.)
In circling area and at AD	TREE	214500.2N 0721024.7E	103 FT	NIL	Group of Trees
In circling area and at AD	POLE	214500.4N 0721040.3E	49 FT	NIL	Light Pole

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TREE	214500.3N 0721034.9E	82 FT	NIL	Group of Trees
In circling area and at AD	POLE	214459.8N 0721033.5E	85 FT	NIL	ELECTRIC POLE
In circling area and at AD	TANK	214459.8N 0721033.3E	87 FT	NIL	Sintex Tank on Building (Building Top 25.9 M.)
In circling area and at AD	BUILDING	214459.7N 0721034.8E	69 FT	NIL	BUILDING
In circling area and at AD	FENCE	214459.6N 0721036.5E	58 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	TREE	214459.3N 0721035.2E	79 FT	NIL	Group of Trees
In circling area and at AD	OTHER	214458.6N 0721039.3E	55 FT	NIL	Mobile Road Traffic (Road Elev. 11.9 M. + 5.0 M. Traffic Ht.)
In circling area and at AD	POLE	214458.2N 0721034.5E	53 FT	NIL	Light Pole
In circling area and at AD	POLE	214457.9N 0721050.2E	46 FT	NIL	Light Pole
In circling area and at AD	FENCE	214457.7N 0721051.5E	49 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	BUILDING	214456.5N 0721014.9E	118 FT	NIL	BUILDING
In circling area and at AD	OTHER	214455.7N 0721036.1E	45 FT	NIL	Approach Light
In circling area and at AD	POLE	214454.9N 0721042.6E	50 FT	NIL	Light Pole
In circling area and at AD	FENCE	214454.3N 0721042.6E	55 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	POLE	214452.2N 0721037.2E	72 FT	NIL	Electric Pole
In circling area and at AD	BUILDING	214443.8N 0721027.3E	102 FT	NIL	BUILDING
In circling area and at AD	TANK	214442.6N 0721022.9E	106 FT	NIL	Sintex Tank on Building (Building Top 30.9 M.)
In circling area and at AD	BUILDING	214442.3N 0721023.4E	99 FT	NIL	BUILDING
In circling area and at AD	POLE	214442.0N 0721022.3E	105 FT	NIL	ELECTRIC POLE
In circling area and at AD	TANK	214442.0N 0721024.4E	100 FT	NIL	Sintex Tank on Building (Building Top 29.3 M.)
In circling area and at AD	TREE	214441.7N 0721024.2E	110 FT	NIL	Group of Trees

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	TANK	214440.6N 0721020.0E	114 FT	NIL	Sintex Tank on Building (Building Top 33.7 M.)
In circling area and at AD	TANK	214440.0N 0721019.0E	115 FT	NIL	Sintex Tank on Building (Building Top 34.2 M.)
In circling area and at AD	SIGN	214515.9N 0721126.1E	25 FT	NIL	Sign Board Arp
In circling area and at AD	OTHER	214519.7N 0721128.5E	34 FT	NIL	GP Hut
In circling area and at AD	NAVAID	214519.8N 0721128.5E	43 FT	NIL	GP DME
In circling area and at AD	NAVAID	214519.9N 0721128.6E	71 FT	NIL	Rod Top GP Main Antenna (Antenna Top 21.4 M.)
In circling area and at AD	NAVAID	214520.9N 0721131.3E	39 FT	NIL	Rod Top GP Monitor Antenna (Antenna Top 11.3 M.)
In circling area and at AD	FENCE	214523.8N 0721136.5E	28 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	OTHER	214522.3N 0721138.5E	36 FT	NIL	Mobile Road Traffic (Road Elev. 6.0 M. + 5.0 M. Traffic Ht.)
In circling area and at AD	FENCE	214522.5N 0721138.7E	28 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	POLE	214522.3N 0721138.9E	25 FT	NIL	Light Pole
In circling area and at AD	POLE	214517.2N 0721140.2E	26 FT	NIL	Light Pole
In circling area and at AD	OTHER	214516.7N 0721139.0E	36 FT	NIL	Mobile Road Traffic (Road Elev. 6.1 M. + 5.0 M. Traffic Ht.)
In circling area and at AD	POLE	214515.9N 0721137.0E	24 FT	NIL	Light Pole
In circling area and at AD	FENCE	214515.7N 0721137.4E	27 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	FENCE	214513.8N 0721136.7E	28 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	OTHER	214514.7N 0721129.8E	23 FT	NIL	PAPI
In circling area and at AD	NAVAID	214508.2N 0721125.4E	54 FT	LGTD	DVOR DME (L)
In circling area and at AD	OTHER	214508.1N 0721124.9E	56 FT	NIL	L/C Rod Near Old DVOR
In circling area and at AD	ANTENNA	214514.4N 0721135.3E	55 FT	LGTD	MET Antenna (L)

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	POLE	214516.4N 0721137.8E	27 FT	NIL	Light Pole
In circling area and at AD	ANTENNA	214519.8N 0721128.1E	39 FT	NIL	RF Antenna on GP Hut
In circling area and at AD	POLE	214523.2N 0721136.5E	26 FT	NIL	Light Pole
In circling area and at AD	FENCE	214505.5N 0721053.4E	44 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	OTHER	214513.2N 0721100.5E	89 FT	NIL	Flood Light Mast
In circling area and at AD	OTHER	214513.9N 0721102.7E	91 FT	NIL	Flood Light Mast
In circling area and at AD	OTHER	214509.2N 0721105.5E	32 FT	NIL	Landing-T
In circling area and at AD	POLE	214513.0N 0721107.2E	42 FT	NIL	POLE
In circling area and at AD	OTHER	214510.3N 0721108.8E	50 FT	NIL	WDI
In circling area and at AD	POLE	214513.8N 0721109.0E	48 FT	NIL	POLE
In circling area and at AD	POLE	214501.4N 0721058.6E	42 FT	NIL	Light Pole
In circling area and at AD	POLE	214502.0N 0721101.4E	52 FT	NIL	POLE
In circling area and at AD	FENCE	214504.0N 0721106.6E	42 FT	NIL	Airport Boundary Wall with Fencing on Top
In circling area and at AD	ANTENNA	214514.0N 0721102.3E	81 FT	LGTD	VHF Antenna
In circling area and at AD	OTHER	214514.0N 0721102.3E	82 FT	NIL	L/C Rod on ATC Tower
In circling area and at AD	TANK	214540.6N 0720837.7E	208 FT	NIL	OHWT
In circling area and at AD	BUILDING	214557.0N 0720846.4E	210 FT	NIL	Temple
In circling area and at AD	TOWER	214610.6N 0720823.1E	410 FT	NIL	Microwave Tower
In circling area and at AD	OTHER	214329.5N 0721030.8E	240 FT	NIL	Mast
In circling area and at AD	STACK	213502.9N 0721319.4E	818 FT	NIL	Chimney (GSECL)
In circling area and at AD	WINDMILL	213620.0N 0720614.3E	1223 FT	NIL	Windmill on Hill
In circling area and at AD	WINDMILL	213625.2N 0720607.0E	1286 FT	NIL	Windmill on Hill (Hill Top 291.9 M.)
In circling area and at AD	WINDMILL	213627.0N 0720628.2E	1240 FT	NIL	Windmill on Hill
In circling area and at AD	NATURAL HIGHPOINT	213629.8N 0720608.4E	975 FT	NIL	Hill Top
In circling area and at AD	NATURAL HIGHPOINT	213631.5N 0720607.1E	987 FT	NIL	Hill Top

In Approach/Take-off/Circling Area and at AD					
1	**2**	**3**	**4**	**5**	**6**
RWY/Area affected	Obstacle type	Coordinates	Elevation (EGM-08)	Marking/LGT	Remarks
In circling area and at AD	WINDMILL	213631.5N 0720607.1E	1312 FT	NIL	Windmill on Hill
In circling area and at AD	WINDMILL	213634.8N 0720601.5E	1252 FT	NIL	Windmill on Hill

VABV AD 2.11 METEOROLOGICAL INFORMATION PROVIDED

1	Name of the associated meteorological office	AMO Ahmedabad
2	Hours of service and, where applicable, the designation of the responsible meteorological office outside these hours	HJ (Sunrise to Sunset). As per requirement of ATC Bhavnagar AMO Ahmedabad
3	Office responsible for preparation of TAFs and periods of validity and interval of issuance of the forecasts	AMO Ahmedabad 09 HOURS
4	Availability of the trend forecast for the aerodrome and interval of issuance	Issued as per ATC Watch Hour when trained staff is available on duty
5	Information on how briefing and/or consultation is provided	Briefing provided as per requistion by user. By receiving from AMO Ahmedabad.
6	Types of flight documentation supplied and language(s) used in flight documentation	As per requirement by user, from OLBS. English
7	Charts and other information displayed or available for briefing or consultation	Not Available
8	Supplementary equipment available for providing information on meteorological conditions, e.g. weather radar and receiver for satellite images;	1. VAISALA FSM (FORWARD SCATTEROMETER) FOR RVR, DCWIS METLOG 4A FOR CURRENT WEATHER DATA AT RWY 25 TDZ. 2. INSTANTANEOUS RVR AVAILABLE FOR RWY 25 TDZ.
9	The air traffic services unit(s) provided with meteorological information	ATS BHAVNAGAR
10	Additional information, e.g. concerning any limitation of service.	HJ (Sunrise to Sunset)

VABV AD 2.12 RUNWAY PHYSICAL CHARACTERISTICS

Designations	TRUE Bearings	Dimensions of RWY (M)	Strength of pavement (PCN) and associated data) and surface of runway and associated stopways	Geographical coordinates for threshold and runway end
1	2	3	4	5
07	67.90 DEG	1921 x 45 M	54/F/C/W/T Asphalt	THR: 214459.34N 0721045.76E RWY END: 214519.66N 0721139.33E
25	247.90 DEG	1921 x 45 M	54/F/C/W/T Asphalt	THR: 214519.66N 0721139.33E RWY END: 214456.16N 0721037.38E

THR elevation and highest elevation of TDZ of precision APP RWY	Slope of runway and associated stopway	Dimensions of stopway (M)	Dimensions of clearway (M)	Dimensions of strips (M)
6	7	8	9	10
THR: 38.0FT TDZ: 38.0FT	-0.34%	NIL	NIL	2041 x 150 M
THR: 22.0FT TDZ: 25.0FT	0.34%	NIL	NIL	2041 x 150 M

Dimensions of runway end safety areas	Location and description of arresting system (if any)	Existence of an obstacle-free zone	Remarks.
11	12	13	14
90M x 90M		NIL	Elevation in EGM08
90M x 90M		NIL	Elevation in EGM08

VABV AD 2.13 DECLARED DISTANCES

RWY Designator	Take-off run available TORA (M)	Take-off distance available TODA (M)	Accelerate distance available ASDA (M)	Landing distance available LDA (M)	Remarks (including runway entry or start point where alternative reduced declared distances have been declared)
1	2	3	4	5	6
07	1921	1921	1921	1661	Nil
25	1851	1851	1851	1851	

VABV AD 2.14 APPROACH AND RUNWAY LIGHTING

Runway Designator	Type, length and intensity of approach lighting system	Runway threshold lights, colour and wing bars	Type of visual slope indicator system	Length of runway touchdown zone lights
1	2	3	4	5
07	SALS 300 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (15.04M)	NIL
25	SALS 180 M LIH	Green	PAPI LEFT/3.00 DEG MEHT (15.36M)	NIL

Length, spacing, colour and intensity of runway centre line lights	Length, spacing, colour and intensity of runway edge lights	Colour of runway end lights and wing bars	Length and colour of stopway lights	Remarks
6	7	8	9	10
NIL	1851 M 60 M White LIH	Red	NIL	Distance of PAPI 07 from RWY 07 THR = 332.09 M
NIL	1851 M 60 M White LIH	Red	NIL	Distance of PAPI 25 from RWY 25 THR = 312.28 M

VABV AD 2.15 OTHER LIGHTING, SECONDARY POWER SUPPLY

1	Location, characteristics and hours of operation of aerodrome beacon/identification beacon (if any)	ABN	At tower building,FLG W & G EV 2 Sec
		IBN	NIL
2	Location and lighting (if any) of anemometer/ landing direction indicator;	LDI	254 DEG, 725M from ARP Unlighted.
		Anemometer	On Tower building. Unlighted.
3	Taxiway edge and taxiway centre line lights;	Edge	NIL
		Centre Line	NIL
4	Secondary power supply including switch-over time;	SECONDARY POWER SUPPLY (02 NOS 320 KVA DG SETS, 01 NO 10 KVA DG SET) AVAILABLE TO ALL THE LIGHTING AT AD. SWITCH OVER TIME LESS THAN 15 SEC. ALSO 02 NOS. 160 KVA UPS SUPPLY AVAILABLE FOR RUNWAY LIGHTING SYSTEM. SWITCH OVER TIME LESS THAN 1 SEC.	
5	Remarks	NIL	

VABV AD 2.16 HELICOPTER LANDING AREA

1	Geographical coordinates of the geometric centre of touchdown and lift-off (TLOF) or of each threshold of final approach and take-off (FATO) area	Not Established
2	TLOF and/or FATO area elevation:	Not Established
3	TLOF and FATO area dimensions to the nearest metre or foot, surface type, bearing strength and marking;	Not Established
4	True bearings of FATO;	Not Established
5	Declared distances available	Not Established
6	Approach and FATO lighting;	Not Established
7	Remarks	Not Established

VABV AD 2.17 AIR TRAFFIC SERVICE AIRSPACE

1	Airspace designation, geographical coordinates and lateral limits	CTR: Circular area centered on 214516N 0721126E within a 10NM radius.
2	Vertical limits	FL 65
3	Airspace classification	D
4	Call sign and language(s) of the air traffic services unit providing service;	Bhavnagar Tower, English
5	Transition altitude	4000 FT
6	Hours of applicability	HO
7	Remarks	Bhavnagar Approach Control Unit is combined with Bhavnagar Aerodrome Control Tower.

VABV AD 2.18 AIR TRAFFIC SERVICES COMMUNICATION FACILITIES

Service Designation	Call sign	Channel(s)	SATVOICE Number(s), if available
1	**2**	**3**	**4**
APP	Bhavnagar Tower	118.150 MHZ	
APP	Bhavnagar Tower	120.800 MHZ	
TWR	Bhavnagar Tower	118.150 MHZ	
TWR	Bhavnagar Tower	120.800 MHZ	
ATIS	----	128.025 MHZ	

Logon address, as appropriate	Hours of operation	Remarks
5	**6**	**7**
	As ATS	1. Functions of Approach Control Service provided by Aerodrome Control Tower. 2. Backup FREQ
	As ATS	1. Functions of Approach Control Service provided by Aerodrome Control Tower. 2. Main FREQ
	As ATS	Back Up
	As ATS	NIL
	As ATS	NIL

VABV AD 2.19 RADIO NAVIGATION AND LANDING AIDS

Type of aids, magnetic variation and type of supported operation for ILS/ MLS, basic GNSS, SBAS and GBAS, and for VOR/ILS/MLS station used for technical lineup of the aid	Identification	Frequency(ies), Channel number(s), Service provider, and reference path identifier(s) (RPI), as appropriate	Hours of operation, as appropriate;
1	2	3	4
LOC 25 CAT I	IBHN	109.900 MHz	As ATS
GP 25		333.800 MHz	As ATS
DME ILS 25	IBHN	CH36X	As ATS
DVOR/DME	BVR	114.100 MHz CH88X	H24

Geographical coordinates of the position of the transmitting antenna	Elevation of transmitting antenna of DME/ elevation of GBAS reference point	Service volume radius from the GBAS reference point	Remarks
5	6	7	8
214454.9N 0721034.1E			
214519.3N 0721126.9E			
214519.3N 0721126.9E	40 FT		1. DME Collocated with GP 2. Geoid Model: EGM-08
214502.3N 0721122.8E	54 FT		1. DME Collocated with DVOR 2. Geoid Model: EGM-08

VABV AD 2.20 LOCAL AERODROME REGULATIONS

1. No aircraft shall make lock turn/ pivotal turn on the Runway to avoid pavement damage.
2. Portion beyond inset RWY fitting END light for RWY25 including turn pad is not available for aircraft operations. ACFT LDG RWY 25 or departure rwy 07 to make 180 degree turn on the Runway.
3. ACFT LDG RWY 07 or departure RWY 25 to make 180 degree turn on turn pad RWY 25 only.
4. RADIO COMMUNICATION FALIURE PROCEDURE

1. INTRODUCTION

Radio communication failure procedures are described in ICAO PANS ATM DOC 4444. Based on these provisions, following radio communication failure procedures are established to standardize the actions to be taken by the pilot of arriving and departing aircraft at Bhavnagar Airport.

The purpose of this document is to establish the standard procedure to handle the event of a radio communication failure situation at Bhavnagar airport to ensure safe and efficient aircraft operation.

2. GENERAL

2.1 As soon as it is known to ATC that two-way communication has failed, the action shall be taken to ascertain whether the aircraft is able to receive transmissions from the air traffic control unit by requesting it to execute a specified manoeuvre (if the aircraft visual with ATC Tower) or to transmit, if possible, a specified signal in order to indicate acknowledgment.

2.2 BHAVNAGAR ATC shall immediately forward information concerning the radio communication failure to all air traffic units concerned along the route of the flight.

2.3 All Transponder-equipped aircraft experiencing Radio Communication failure shall set transponder to Mode A/C code 7600 as soon as practicable.

2.4 Immediately after the detection of RCF, the aircraft shall attempt to establish communications with the appropriate air traffic control unit using all other available means. In addition, the aircraft shall keep a watch for any instructions that may be issued by visual signals.

2.5 The pilot shall make blind transmission to ATC of all necessary reports and actions taken by the aircraft, e.g. descent, turn, proceeding to waypoint, etc., irrespective of whether partial/ complete RCF has been established or not.

2.6 Aircraft experiencing radio communication failure shall be provided all required information necessary for the safe and efficient conduct of flight and landing clearance without expecting acknowledgment.

3. ASSIGNED RUNWAY AND ITS AVAILABILITY FOR RCF AIRCRAFT:

3.1 In case of an arriving aircraft experiencing RCF, when the Runway for landing has already been advised to the aircraft by BHAVNAGAR ATC via VHF, such runway shall be considered as the assigned runway. If an arriving aircraft has not been advised of any runway, Runway 25 shall be considered as an assigned runway for such arrival.

NOTE: In the case of RCF, pilots shall not assume runway on the basis of ATIS (128.025 MHz) Broadcast.

3.2 In case of departures from Bhavnagar airport is returning due to RCF, Runway 25 shall be considered as the assigned runway for landing, irrespective of the runway used fordeparture.

3.3 Runway lights and Approach lights shall be in the ‘SWITCHED ON’ Position to indicate the availability of such Runway for aircraft experiencing RCF irrespective of visibility and Day or Night.

3.4 Runway and Approach lights in the ‘SWITCHED OFF’ position shall Indicate the non availability of such Runway for aircraft experiencing RCF.

3.5 In case strong tailwind conditions are encountered during approach for the Runway as mentioned in para 3.1, the aircraft experiencing RCF will carry out a missed approach on the runway as mentioned in para 3.1 and after following the complete missed approach procedure for such approach, aircraft will carry out the published approach rocedure for the opposite runway

4. PROCEDURE FOR ARRIVALS:

4.1 In case radio communication failure takes place after establishing the ILS/final approach track, aircraft may continue the approach and land if visual, or go around and carry out the missed approach procedure and join the VOR (114.100) BVR holding procedure climbing to and maintaining 2400 ft.

4.2 Except when descent clearance has already been received from ATC BHAVNAGAR and acknowledged, the pilot shall continue on the ATS route maintaining filed fligh plan level or level cleared by adjacent ATC/ ACC Unit i.e. VASU ATC, VABB AREA or VAAH AREA and do not commence descent before 25NM VOR (114.100) BVR.

4.3 After crossing 25NM, the pilot shall descend to FL080 to join the published hold over VOR (114.100) BVR for the runway as mentioned in para 3.1.

4.4 Commence descent to 2100 feet in published hold over VOR (114.100) BVR at, or as close as possible to, the expected approach time (EAT) last received and acknowledged;or, if no expected approach time (EAT) has been received and acknowledged, at, or as close as possible to, the estimated time of arrival (ETA) over VOR (114.100) BVR.

4.5 Leave VOR (114.100) BVR and carry out the published ILS/VOR Procedure for the runway as mentioned in para 3.1.

4.6 Land, if possible, within 30 minutes after the estimated time of arrival (ETA) or the last acknowledged expected approach time (EAT), whichever is later.

5. PROCEDURE FOR DEPARTURE:

5.1 Departing aircraft experiencing radio communication failure and intending to continue to its filed flight plan destination shall:

5.1.1 Continue ATS route as per filed flight plan, climbing to/maintaining cleared and acknowledged flight level, or minimum flight altitude if higher, for a period of 20 minutes following the aircraft’s failure to report its position over compulsory reporting point and thereafter adjust level and speed in accordance with the filed flight plan to destination.

5.1.2 When the aircraft is given clearance for offset from flight plan route in without a specified limit, re-join the current flight plan route after passing 25 NM from VOR (114.100) BVR but no later than the next significant point, taking intoonsiderationthe applicable minimum flight altitude.

5.2 A departing aircraft, experiencing radio communication failure and intending to land back at Bhavnagar Airport, shall:

5.2.1 Maintain the last assigned level, or minimum flight altitude if higher, for a period of 10 minutes on the flight plan route or on offset radial as cleared in ATC clearance.

5.2.2 After 10 minutes turn towards VOR (114.100) BVR via the same flight plan route or radial as per departure clearance. On Reaching VOR (114.100) BVR joins hold as published.

5.2.3 If required jettison fuel, then remain in hold over VOR (114.100) BVR after taking all precautions.

5.2.4 Commence descent in the hold to 2100 feet. Leave VOR (114.100) BVR and carry out the published ILS/VOR Procedure for the runway as mentioned in para 3.2.

5.3 Procedures for VFR Traffic:

5.3.1 In case of a VFR flight experiencing RCF and does not intend to land at Bhavnagar:

It shall continue to fly in VMC, land at the nearest suitable aerodrome/ helipad, and report its arrival by the most expeditious means to the ATC, Bhavnagar.

5.3.2 In case of a VFR flight experiencing RCF and intending to land at Bhavnagar:

a) Proceed according to the current flight plan route in VMC and maintain the altitude/level last assigned and acknowledged.

b) Proceed towards 15 NM North of the field subject to arrival route and descent

1000 FT AGL.

c) Maintaining 1000 FT AGL proceed to Bhavnagar to join the traffic circuit for the last acknowledged Runway or the Runway as per para 3.1.

d) Carry out a complete aerodrome traffic circuit and land on the last acknowledged Runway or the Runway as per para 3.1.

• When Bhavnagar ATC becomes aware of the RCF of the aircraft, the controller concerned shall accord priority for the RCF aircraft to land.

• If the aircraft re-establishes communication or has landed, the ATC unit shall inform all concerned units/aerodromes along the route as per the flight plan including the alternate aerodrome.

5.4 Various Signals used by ATCO during communication failure Air traffic controllers have ALDIS lamps as backup systems for communication if the radio fails.

The various signals used by the ATCO during communication failure are as follows	**Aircraft in Flight**	**Aircraft on the ground**
Steady green	Cleared to land	Cleared for take-off
Steady red	Give way to other aircraft, continue circling	Stop
A series of green flashes	Return for landing* (*Clearances to land and to taxi will be given in due course.)	Cleared to taxi
Series of red flashes	The aerodrome is unsafe, do not land	Taxi clear of landing area in use
Series of white flashes	Land at this aerodrome and proceed to the apron*	Return to starting point on the aerodrome
Alternate Red and Green	Exercise extreme caution	Exercise extreme caution
Blinking runway lights	Vehicles, Plans, and pedestrians immediately clear the landing area in use	

LOW VISIBILITY PROCEDURES

1.INTRODUCTION

1.1 Until the latest amendment of DGCA Civil Aviation Requirements (CAR) section 8, Series 'C', Part-I on All-Weather Operations, Low visibility procedures were required at aerodromes for the purpose of ensuring safe operations during Categories II and III approaches and/or low visibility take-offs (LVTO).

1.2 However, in latest amendment to CAR (Rev. 10, issue Oct 2022), para 5.3, following provision regarding Low visibility procedures is added. "An operator shall not conduct takeoff with RVR/visibility less than standard Category I conditions of 550 m RVR/800 m visibility unless low visibility procedures are enforced".

1.3 Further, as per para 5.2 of DGCA CAR Section 8, Series C, Part-1 on All Weather Operations, "An operator shall not conduct low visibility take-offs in less than 400 m RVR unless approved by DGCA. Operators may be authorized LVTO minima of up to 75m".

1.4 These provisions necessitated the need for low visibility procedures for accommodating/permitting departures in Visibility/RVR less than 800M/550M even at such airports where there are no CAT II and CAT III operations

1.5 As per provisions of Aerodrome Design and Operations CAR and ICAO Annex 14, Runway centre line lights are required for take-off in RVR below 400 M. Further, as per Aerodrome Design and operations CAR para 9.8.7, Surface Movement Radar (SMR) need to be provided at an aerodrome intended for use in Runway Visual Range conditions less than a value of 350 M.

1.6 Accordingly, as per the provisions of DGCA CAR Section 8, Series C, Part-1 on All Weather Operations; DGCA CAR Section 4, Series B, Part-1 on Aerodrome Design and Operations; ICAO DOC 9365; DGCA Aerodrome Advisory Circular AD AC NO. 4 of 2022; and DGCA letter no. AV1503/02/03-AL dated 12.01.2018; Low visibility Procedures have been developed for BHAVNAGAR Airport to accommodate/permit departures in Visibility/RVR less than 800M/550M up to 400 M RVR from RWY 25.

2. PURPOSE

2.1 The purpose of this document is to define the Standard Operating Procedures (SOP) for Low Visibility Procedures at Bhavnagar Airport.

3. SCOPE

3.1 This is applicable to all the concerned personnel involved in the Low Visibility Procedures as mentioned in this SOP.

3.2 The Low Visibility Procedure (LVP) describe the procedures and actions that are required to be taken by the Air Traffic Control (ATC), Pilots, Airline Operators, MET dept, Vehicle Operators, Airport Operations, Electrical Engg., CNS, Airport Fire Services and Ground Handling Agencies during Low Visibility procedures - Take-Off at Bhavnagar Airport.

3.3 The procedures contained in this document shall be read in conjunction with other applicable ICAO DOCs and Annexes / DGCA CARs and relevant circulars, AAI CHQ/RHQ instructions on the subject issued from time to time.

4. DEFINITIONS

4.1 Aerodrome Operating Minima:

The limits of usability of an aerodrome for:

i) take-off, expressed in terms of runway visual range and/or visibility and, if necessary, cloud conditions.

ii) landing in 2D instrument approach operations, expressed in terms of visibility and/or runway visual range; minimum descent altitude/height (MDA/H) and, if necessary, cloud conditions; and

iii) landing in 3D instrument approach operations, expressed in terms of visibility and/or runway visual range and decision altitude/height (DA/H) appropriate to the type and/or category of the operation.

4.2 Instrument approach operations (Annex 6): An approach and landing using instruments for navigation guidance based on an instrument approach procedure. There are two methods for executing instrument approach operations:

i) a two-dimensional (2D) instrument approach operation, using lateral navigation guidance only; and

ii) a three-dimensional (3D) instrument approach operation, using both lateral and vertical navigation guidance

Note – Lateral and vertical navigation guidance refers to the guidance provided either by:

i) a ground-based radio navigation aid; or

ii) computer-generated navigation data from ground-based, space-based, self-contained navigation aids or a combination of these.

Instrument approach operations shall be classified based on the designed lowest operating minima below which an approach operation shall only be continued with the required visual reference as follows:

i) Type A: a minimum descent height or decision height at or above 75 m (250 ft); and

ii) Type B: a decision height below 75 m (250 ft). Type B instrument approach operations are categorized as:

a) Category I (CAT I): a decision height not lower than 60 m (200 ft) and with either a visibility not less than 800 m or a runway visual range not less than 550 m;

b) Category II (CAT II): a decision height lower than 60 m (200 ft) but not lower than 30 m (100 ft) and a runway visual range not less than 300 m;

c) Category III (CAT III): a decision height lower than 30 m (100 ft) or no decision height and a runway visual range less than 300 m or no runway visual range limitations.

4.3 Decision Altitude/Height: A specified altitude or height in a 3D instrument approach operation at which a missed approach must be initiated if the required visual reference to continue the approach has not been established.
Note: Decision altitude (DA) is referenced to mean sea level and decision height (DH) is referenced to the threshold elevation.
4.4 ILS Critical Area: An area of defined dimensions about the localizer and glide path antennas where aircraft and vehicles are excluded during all ILS operations. The critical area is protected because the presence of vehicles / or aircraft inside its boundaries will cause unacceptable disturbance to the ILS signal-in-space.
4.5 ILS Sensitive Area: An area extending beyond the ILS critical area where the parking and/or movement of vehicles, including aircraft, are controlled to prevent the possibility of unacceptable interference to the ILS signal during ILS operations. The sensitive area is protected to provide protection against interferences cause by large moving objects outside the critical area but still normally within the airfield boundary.
4.6 Low Visibility Procedures: Specific procedures applied at an aerodrome for the purpose of ensuring safe operations during Categories II/ III approaches and/or low visibility take-offs.
Note: As per para 5.3 of CAR on All Weather Operations, an operator shall not conduct Take-off with RVR/Visibility less than standard CAT-I conditions of 550m RVR/800m Visibility unless low visibility procedures are enforced.
4.7 Low Visibility Take-Off (LVTO): A term used in relation to flight operations referring to a take-off on a Runway where the RVR is less than 400m.
4.8 Manoeuvring Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, excluding aprons.
4.9 Movement Area: That part of an aerodrome to be used for the take-off, landing and taxiing of aircraft, consisting of the manoeuvring area and the apron(s).
4.10 Runway Visual Range: The range over which the pilot of an aircraft on the centre line of a runway can see the runway surface markings or the lights delineating the runway or identifying its centre line.
4.11 Visibility (Doc 9365): Visibility for aeronautical purposes is the greater of:
i) the greatest distance at which a black object of suitable dimensions, situated near the ground, can be seen and recognized when observed against a bright background;
ii) the greatest distance at which lights in the vicinity of 1000 candelas can be seen and identified against an unlit background.
Note 1: The two distances have different values in air of a given extinction coefficient, and the latter ii) varies with the background illumination. The former i) is represented by the meteorological optical range (MOR).
Note 2: The definition applies to the observations of visibility in local routine and special reports, to the observations of prevailing and minimum visibility reported in METAR and SPECI and to the observations of ground visibility.

5. ABBREVIATIONS

5.1The abbreviations used in descriptions of Low Visibility Procedures have the following meanings:
ARFF: AIRCRAFT RESCUE & FIRE FIGHTING
AGL: AERONAUTICAL GROUND LIGHT
AOM: AERODROME OPERATING MINIMA
ATC: AIR TRAFFIC CONTROL
ATIS: AUTOMATIC TERMINAL INFORMATION SERVICE
CFT: CRASH FIRE TENDER
DG: DIESEL GENERATING SET
GP: GLIDE PATH
ILS: INSTRUMENT LANDING SYSTEM
IHP: INTERMEDIATE HOLDING POSITION
LOC: LOCALIZER
LSA: LOCALIZER SENSITIVE AREA
LVP: LOW VISIBILITY PROCEDURE
MID: MID POINT
MET: METEOROLOGY
NAV: NAVIGATIONAL
RVR: RUNWAY VISUAL RANGE
SMC: SURFACE MOVEMENT CONTROL
SP: SAFEGUARDING PROCEDURES
SSO: SHIFT SUPERVISORY OFFICER
TDZ: TOUCHDOWN ZONE
TSO: TOWER SUPERVISORY OFFICER
WSO: WATCH SUPERVISORY OFFICER

6. GENERAL

6.1 The Low Visibility Procedure (LVP) incorporates safeguarding measures to mitigate runway incursions and defines operational restrictions to ensure safe Airside Operations taking into account the available Aerodrome facilities.

7. MINIMUM REQUIREMENTS:

7.1 The following Aeronautical Ground lights and RVR equipment shall be serviceable to the required standard to support Low Visibility Procedures.

i) Runway edge lights,

ii) Runway end lights,

iii) Instrumented RVR at the beginning (TDZ RVR).

iv) Stand by Power supply to maintain switch over time of 1 Second for Runway Edge Lights and Runway End Lights. This requirement can be met with the help of DG Set and/or UPS.

7.2 Unserviceability of Aeronautical Ground Lights/ Equipment before Implementation of LVP

Low Visibility Procedures will not be implemented when any of the light/equipment mentioned in para 7.1 above is un-serviceable or is not maintained as per the required standard.

Aeronautical Ground Lighting Facility	Un-serviceability	Restrictions
Runway Edge lights	More than 15% of all lights are unserviceable	LVP operations will be suspended.
	Any two consecutive lights or more are unserviceable	
Runway End lights	More than 15% of all lights are unserviceable	LVP operations will be suspended
	Any two consecutive lights or more are unserviceable	
Standby Generators/UPS	Any of the generator/UPS is unserviceable	LVP operations will be suspended.
RVR system	RVR system unserviceable	LVP operations will be suspended.

7.3 Unserviceability of Aeronautical Ground Lights / Equipment after Implementation of LVP:

When any of the light/equipment mentioned in para 7.1 above becomes un-serviceable or fails to meet the required standard during periods of LVP, TWR shall advise the aircraft accordingly and LVP shall be suspended and information to this effect shall be included in ATIS broadcast.

8.SAFEGUARDING PROCEDURES

8.1 Safeguarding Procedures (SP) are instructions for relevant airport departments to prepare ground services and facilities for low visibility operations in order that when LVP are implemented all safeguarding procedures are complete. WSO/Duty Officer Tower will initiate and co-coordinate with all the concerned agencies for completion of safeguarding procedures before implementation of Low Visibility Procedures.

8.2 Safeguarding Procedures shall be initiated when:

i) The Visibility/RVR is less than 1200m and visibility/RVR is forecast to deteriorate to 800 m or less; and/or

ii) The cloud ceiling is less than 400ft and forecast to fall to 200ft or less.

8.3 Whenever meteorological conditions stipulated in para 8.2 prevails or anticipated WSO/Duty Officer Tower shall coordinate with Airport Rescue Fire Station (ARFS) for implementing the Safeguarding Procedures (SP). WSO/Duty Officer Tower shall also coordinate with following agencies for initiation of implementation/ cancellation/completion of SP as appropriate.

8.3.1 Coordination responsibilities are divided as follows:

i) ATC will inform:

a) Airport Rescue Fire Station (ARFS)

b) CNS/Equipment Room

c) Duty MET Officer

d) Airside Operations/Management (ASM)

ii) Airside Operations (ASM) will inform:

a) Concerned Airlines

b) Terminal Manager

c) Electrical Engineering/Powerhouse

d) M. T. Section

e) Civil Engineering

f) CISF

g) BPCL

8.4 The ASM will inform the above-listed agencies (Airlines, BPCL, Electrical, Civil, M.T., T.M. and CISF) and, upon confirmation of the implementation of safeguarding procedures, will confirm it to ATC.

8.5 When all the concerned agencies have completed their necessary actions, they shall report to WSO/Duty Officer Tower that their Safeguarding Procedure (SP) is completed and the airport is safeguarded for LVP operations.

8.6 Actions to be taken by Various Agencies:

8.6.1 Before commencement of winter season, a meeting will be held by Airport Director in the month of October every year to inform all airlines and agencies operating at airport about their roles/ responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.

8.6.2 A refresher program for ATCO's and personnel responsible for airside operations should be conducted every year.

8.6.3 All the agencies shall ensure that staff and drivers are suitably trained during Low visibility procedure operations.

8.7 WSO/Duty Officer Tower shall ensure that the provision mentioned in para 8.7 are complied in respect of SP and proper log entry shall be made in respect of initiation of implementation/ cancellation/completion of SP as appropriate.

8.8 Safeguarding procedures include:

i) Positioning of CFTs at the predetermined position [PDP].

ii) Stopping of all maintenance works on the manoeuvring area as well as removal of all men and mobile equipment from the said area.

iii) Ensuring availability of secondary power supply for change over time of maximum one second for RWY Edge and RWY End lights supported by UPS.

RWY Edge and RWY End lights may continue to operate on main power supply during safeguarding Procedures. Whenever, LVP is to be implemented as per para 9 below, the RWY Edge and RWY End lights shall be put on Standby Power Supply (DG set or UPS). This operation needs to be completed before LVP is implemented.

As UPS is available at Bhavnagar Airport and is capable of maintaining the required AGL system (refer table under para 7.2) with one second of Switch Over time with Main Supply, the main supply can continue to be primary supply and the Generator Supply can be kept as Standby Power supply. In case of UPS is unserviceable, Generator supply will become primary source of power supply and Main power supply shall act as standby power supply.

iv) Ensuring that Runways, Taxiways and other associated facilities such as Markings, Aeronautical Ground Lights are inspected within one hour preceding the implementation of LVP Operations. This is done to ensure that the lighting systems and markings are serviceable as described. Thereafter, Inspections should be done every two hours while LVP is in progress. These lighting inspections should be accorded priority and, if necessary, aircraft operations may be delayed.
v) Ensure all access roads are closed.

9. LOW VISIBILITY PROCEDURES-TAKE OFF

Implementation of Low Visibility Take off Procedures:

MET officer shall inform WSO/Duty Officer Tower controller whenever Visibility/RVR reduces to 800 Meters or below and/ or cloud ceiling is at 200 ft or below. WSO/Duty officer Tower controller shall coordinate with all the agencies (as mentioned in 8.3.2 above) to confirm whether the Safeguarding procedures have been completed or not. When Visibility/ RVR falls below 800M/550M and or Cloud Ceiling is 200 ft or below and safeguarding procedures are complete (as mentioned in 8.4 and 8.5) WSO/Duty Officer Tower will implement Low Visibility Procedures. WSO/Duty Officer Tower shall inform all users of the imposition of low visibility take off procedures (as stipulated in 8.3.1 above).

9.1 A take-off alternate aerodrome shall be selected and specified in the operational flight plan if either the meteorological conditions at the aerodrome of departure are below the operator's established aerodrome landing minima for that operation or if it would not be possible to return to the aerodrome of departure for other reasons.

9.2 Take-off minima established by the operator must be expressed as visibility or RVR limits, considering all relevant factors for each aerodrome planned to be used and the aeroplane characteristics.

9.3 RVR/Visibility for Take-off (Commercial Transport Aeroplanes) is as per Table, below:

Take-off RVR/Visibility	
Take-off RVR/Visibility	RVR/VIS[1] CAT A, B, C & D
Adequate Visual reference[2] (Day only)	500 m
Runway edge lights or Runway Centre line markings[3]	400m
Runway edge lights and Runway Centre line markings[3]	300m
Runway edge lights and Runway Centre line lights	200m
Runway edge lights and Runway Centre line lights and relevant RVR information[4]	150m
High intensity Runway edge lights and Runway Centre line lights (spacing 15 m or less) and relevant RVR information[4]	125m
High intensity Runway edge lights and Runway Centre line lights (spacing 15 m or less), approved lateral guidance system and relevant RVR information[4]	75m

Note 1 – The TDZ RVR/VIS may be assessed by the pilot subject to Pilot fulfilling conditions of para 14,15 & 16 of DGCA CAR Section 8 Series C Part –I Annexure 2.

Note-1a: Pilot assessment of TDZ RVR is to be made only when the MID & ROLL OUT zone RVR are reported and both these are not less than 200 m. (DGCA CAR Section 8 Series C Part –I Annexure 2 Para 17)

Note1b: An operator shall not conduct take-off with RVR/visibility less than standard Category I conditions of 550m RVR/ 800 m visibility unless low visibility procedures are enforced. (DGCA CAR Section 8 Series C Part –I Para 5.3)

Note1c: Below 200 m all three RVR are required. The governing RVR shall be the lowest of the reported RVRs. (DGCA CAR Section 8 Series C Part –I Table 10: RVR/Visibility for take-off).

Note 2 - Adequate Visual reference means, that a pilot is able to continuously identify the take-off surface and maintain directional control.

Note 3 - For night operations at least runway edge lights or centre line lights and runway end lights are available.

Note-4: The required RVR must be achieved for all relevant RVR reporting points (touchdown, mid-point and stop-end/ rollout). The governing RVR shall be the lowest of the reported RVRs.

Note-5: An operator shall not conduct low visibility take-offs in less than 400m RVR unless approved by DGCA.

10.ACTIONS BY VARIOUS UNITS DURING LVP

10.1 Actions by ATC:

i) Inform CNS, Aircraft Rescue & Fire Fighting Services, Airside operations, MET, etc.
ii) Only one aircraft shall be permitted at a time on the maneuvering area during the time LVP is in force.
iii) Towing of aircraft during LVP will be permitted only in case of extreme operational requirement. TWR shall ensure that the towing of aircraft is done under escort of "Follow Me" vehicles. "Follow Me" shall follow the route cleared by ATC.
iv) TWR shall not permit any ground run on the manoeuvring area except idle power run on the stands;
v) TWR shall ensure that "Follow Me" services are provided to pilots on request;
vi) The number of the vehicles on the manoeuvring area shall be restricted to bare minimum and records of all vehicles operating on the manoeuvring area shall be maintained and all vehicles operating on the manoeuvring area shall be in two-way contact with Aerodrome Control Tower.
vii) The following may be included in ATIS: "LOW VISIBILITY PROCEDURES IN FORCE".
viii)TWR shall permit departures only from the beginning of the Runway in use.
ix) Whenever visibility/RVR is less than 800M/550M, Duty officer tower shall confirm from pilot that the reported RVR value is within minima before issuing take-off clearance.
x) Duty Officer Tower shall stop all flight operations whenever RVR falls below 350 M.
xi) Runway inspection should be done when interval between two aircraft movements is 30 minutes or more.
xii) The bird sarers deployed in the operational area shall be specifically instructed not to come near the Runway/Taxiway.
xiii) Inform changes in RVR readings to the departing aircraft as required.
xiv) Inform pilots of all failures of ILS, lighting system etc. relevant to Low Visibility Operations.
xv) Initiate emergency action if aircraft on CAT I ILS is not in radio contact, as expected.
xvi) Duty officer tower shall take the Take – off Alternate from pilot before issuing take-off clearance.

10.2 Actions by CNS In-charge/ Duty Officer, NAV AIDS:

i) On receipt of "Initiating SP" from ATC, CNS In-charge will inform the Duty Officer, NAV AIDS and have the ILS equipment and its status indicators in ATC units checked up. He will inform ATC of any unserviceability in the equipment which is likely to affect ILS CAT I operation.
ii)On receipt of "Initiating LVP" from ATC that LVP are to be made effective CNS In-charge will maintain continuous watch on the performance of ILS equipment and will inform ATC of any un-serviceability which may affect ILS CAT I operation.

10.3Actions by Duty Manager, Airside operations:

Duty officer shall ensure that:
i)No vehicle/person enters or is present in the sensitive/critical areas of localizer and glide path.
ii)Inform CISF, Airlines, T.M., M.T., Electrical, Civil & BPCL, etc.
iii)All civil/electrical works in progress are to be stopped in the manoeuvring area immediately and men/material/ equipment to be removed from the sensitive/critical areas of localizer and glide path.
iv)Where low visibility procedures are in effect, persons and vehicles operating on an apron shall be restricted to the essential minimum.
NOTE:After ensuring above, Follow-me vehicles/operational vehicles will confirm the same to the ATC.
Subsequently, follow-me vehicle/ operational vehicles shall remain available and will maintain listening watch on Walkie-Talkie.

Any information about unserviceability of any of the Runway visual aids or power supply system will be immediately informed to ATC Tower accordingly.
No vehicles shall enter/cross in the vicinity of Runway without permission from ATC Control Tower. Such vehicles shall be able to communicate with ATC.
The vehicles of the ARFF services, civil, electrical division or of any other agency which are not equipped with RT but has to enter the Runway or taxiway for urgent operational requirement shall be escorted by the follow me jeep all the time.
10.4 Action by Civil Incharge:
i) Shift Manager (Civil) to ensure that all civil works in progress in manoeuvring area, are stopped and that the work area is restored in complete serviceable condition and confirm to WSO/Duty Officer ATC (Through ASM) accordingly.
ii) Shift Manager, Civil shall ensure closure of roads leading to critical/sensitive area by chains.
iii) During CAT I operations, ATC may authorize operations of vehicles Specially trained about
iv) Critical and sensitive area, on manoeuvring area, including the crossing of taxiways. Ensure that Runways, Taxiways and other associated facilities such as Markings, Lights are inspected every 2 hours while LVP is in progress.
10.5 Action by Electrical Incharge:
i) On receipt of advice to implement Low Visibility Procedures from ATC, Shift Manager (Electrical) will check that following visual aids are serviceable and can be operated at full intensity:
a) Approach lighting system
b) Approach supplementary lighting
c) Runway edge lights
d) Runway threshold and end lights
NOTE 1: The Electrical Department Personnel shall be present in Control Tower during periods of reduced visibility and ensure that the ALCMS display(s) in the Control Tower are serviceable.
NOTE 2: No adjustment in light intensity shall be made without permission from ATC Tower during LVP.
ii) They shall ensure that no electrical maintenance works is carried out during LVP either in powerhouse or on any other electrical facilities used during CAT I operations.
iii) They will inform the un-serviceability or any change in status of any facility/systems to ATC & Airside Ops Manager immediately.

10.6 Actions by ARFF:

i) Positioning of crash fire tenders on pre-determined position (PDP).
ii) ARFF/ CFTs must obtain clearances from ATC prior to entry to any aircraft movement area other than the designated service roads.

10.7 Actions by CISF:

i) The officer in charge on duty shall ensure by deploying adequate manpower that vital electrical installations are properly secured and protected against any unauthorized intrusion.
ii) The Officer in charge, on receipt of advice to implement Low Visibility Procedures, will immediately inform all access gates and CISF posts under their respective controls in operational area to restrict movement of vehicles under their control.

10.8 Actions by Duty Officer (Meteorological Office):

i) Duty Meteorological Officer would issue an 'Outlook for Low Visibility Procedures' to the ATC whenever he expects that the RVR and/or cloud ceiling will fall below 800 m and/or 400ft or less respectively.
ii) Whenever Duty Meteorological Officer visualizes that RVR is likely to fall below 800 m and/or cloud ceiling to 200ft or less within next 2 hours, he will issue an 'Advisory Message' to ATC to this effect.
iii) When the RVR and/or cloud ceiling are 800m and/or 200ft respectively and the trend is towards improvement in these elements of weather conditions the Duty Met Officer may, when requested by ATC, advise him about such improving weather conditions for the purpose of termination of LVP.
iv) The Duty Met Officer shall be present in Control Tower during periods of reduced visibility and ensure that the RVR displays in the Control Tower is serviceable.
v) In case, Instrumented RVR is not available, real time RVR shall be provided, whenever requested.

11. TERMINATION OF LOW VISIBILITY TAKE OFF PROCEDURES

11.1 When Visibility/RVR improves to 800M/550 M or more and cloud ceiling is 200 feet or higher and trend is for improvement, WSO/Duty Officer Tower would terminate operations of LVP-Take Off. He may obtain advice from Duty Met. Officer regarding improvement in weather conditions before the termination of LVP.
i) The WSO/Duty Officer Tower will intimate the following regarding termination of operations:
a) ARFF
b) Duty Met Officer
c) CNS/Equipment Room
d) Airside Operations
ii) ASM will intimate all previously notified personnel (undermentioned) to resume normal operations:
a)Terminal Manager
b)Civil & Electrical Engg.
c)Airlines

d)CISF
e)BPCL
11.2 On cancelling of LVP, following message shall be included in two subsequent ATIS broadcasts. “LOW VISIBILITY PROCEDURES CANCELLED”.
11.3 If SP are implemented and LVP are not subsequently implemented and the visibility/RVR improves and is more than 1200 m and/or the cloud ceiling is 400ft or higher and both are forecast to remain above the required SP criteria, WSO/Duty Officer Tower may cancel SP.

12. ACTIONS BY OTHER AGENCIES (AIRLINES, REFUELLING COMPANIES, CATERING AGENCIES, ETC.)

12.1 Every year before commencement of monsoon/winter season, a meeting will be held by Airport Director, to inform all airlines and agencies operating at airport about their roles/responsibilities and create awareness to ensure cooperation for safe airport operations during periods of low visibility.
12.2 All the agencies shall ensure that staff and drivers are suitably trained during Low Visibility Procedure operations.
12.3 All agencies operating in the operational area shall ensure that only those vehicles that are absolutely essential for aircraft operations operate in the operational area during periods of low visibility. The drivers of these vehicles should keep a look out for taxiing aircraft and other vehicles to prevent accidents.
12.4 All the vehicles must have their obstruction lights “ON” during Low Visibility Procedures operations.
12.5 All instructions/sign boards provided for vehicular movement area/service roads must be followed while operating in the operational area

VABV AD 2.21 NOISE ABATEMENT PROCEDURES

NIL

VABV AD 2.22 FLIGHT PROCEDURES

NIL

VABV AD 2.23 ADDITIONAL INFORMATION

1.Bird activity in and around airfield. Pilots to exercise caution.

2.Additional caution for hills with respect to ARP Bhavnagar:

i. Hill Top: 994FT 210 DEG 10NM
ii. Hill Top: 541FT 235 DEG 12NM
iii. Hill Top: 610FT 258 DEG 12NM

3. **Exemption Granted by DGCA:**
DGCA India has granted exemption vide letter dated 11.10.2024 for non-compliant width of runway strip for RWY 07/25 against provisions in CAR Section 4 Series B Part I Para 3.4.3 for the period up to 31.07.2026.
4. Local FLY in progress beyond ATC watch in and around VABV by FLY TRG organization.All ACFT transiting through VABV are advised to EXER CTN for TFC and CK TFC PSN. FLY TRG organization [dunes aviation] has been allotted FREQ 131.850MHZ.Contact details of GND is +916353182899.
5. Heavy kite FLY ACT are there in and around APCH specially on APCH RWY 07 BTN months of NOV and FEB every year. PIlots To EXER CTN while LDG and TKOF.
6. Pilot EXER CTN while LDG/TKOF due jackal menace.
7. Turn pad for DEP RWY 25 is available for Code "C" Aircraft B737 (700).
PCN value for Turn Pad is: 54/F/C/W/T
Surface: Asphalt

VABV AD 2.24 CHARTS RELATED TO AN AERODROME

1 Aerodrome Chart (12 June 2025)

2 Aircraft Parking/Docking Chart (12 June 2025)

3 Aerodrome Obstacle Chart Type A (Operating Limitations) RWY 07(17 April 2025)

4 Aerodrome Obstacle Chart Type A (Operating Limitations) RWY 25(17 April 2025)

5 ILS Procedure RWY 25

6 VOR Procedure RWY 07 (17 APR 2025)

7 Instrument Approach Chart - VOR Procedure RWY 25 (CAT A/B/C) (08 August 2024)

8 RNP Y Procedure RWY 25

9 Coding Table - RNP Y Procedure RWY 25

VABV AD 2.25 VISUAL SEGMENT SURFACE (VSS) PENETRATION

Procedure	Procedure Minima	VSS Penetration
1	2	3